Example 5.1

Find the unknowns in the following figures

(i)

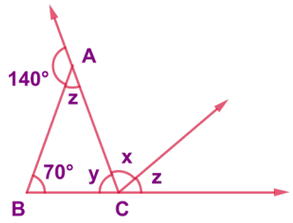


Fig. 5.7(i)

(ii)

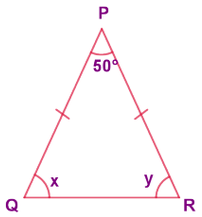


Fig. 5.7(ii)

(iii)

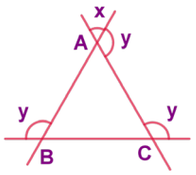


Fig. 5.7(iii)

Solution:

(i) Now, from Fig. 5.7(i), $140° + z = 180°$ (linear pair)

$$\Rightarrow z = 180° - 140° = 40°$$

Also $x + z = 70° + z$ (exterior angle property)

$$\Rightarrow x = 70°$$

Also $z + y + 70° = 180°$ (angle sum property in ΔABC)

$$\Rightarrow 40° + y + 70° = 180°$$

$$\Rightarrow y = 180° - 110° = 70°$$



(ii) Now, from Fig. 5.7(ii), $PQ = PR$

$$\Rightarrow \angle Q = \angle R$$ (angles opposite to equal sides are equal)

$$\Rightarrow x = y$$

$$\Rightarrow x + y + 50° = 180°$$ (angle sum property in ΔPQR)

$$\Rightarrow 2x = 130°$$

$$\Rightarrow x = 65°$$

$$\Rightarrow y = 65°$$

(iii) Now, from Fig. 5.7(iii), in ΔABC $\angle A = x$ (vertically opposite angles)

Similarly $\angle B = \angle C = x$ (Why?)

$$\Rightarrow \angle A + \angle B + \angle C = 180°$$ (angle sum property in ΔABC)

$$\Rightarrow 3x = 180°$$

$$\Rightarrow x = 60°$$

$$\Rightarrow y = 180° - 60° = 120°$$

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